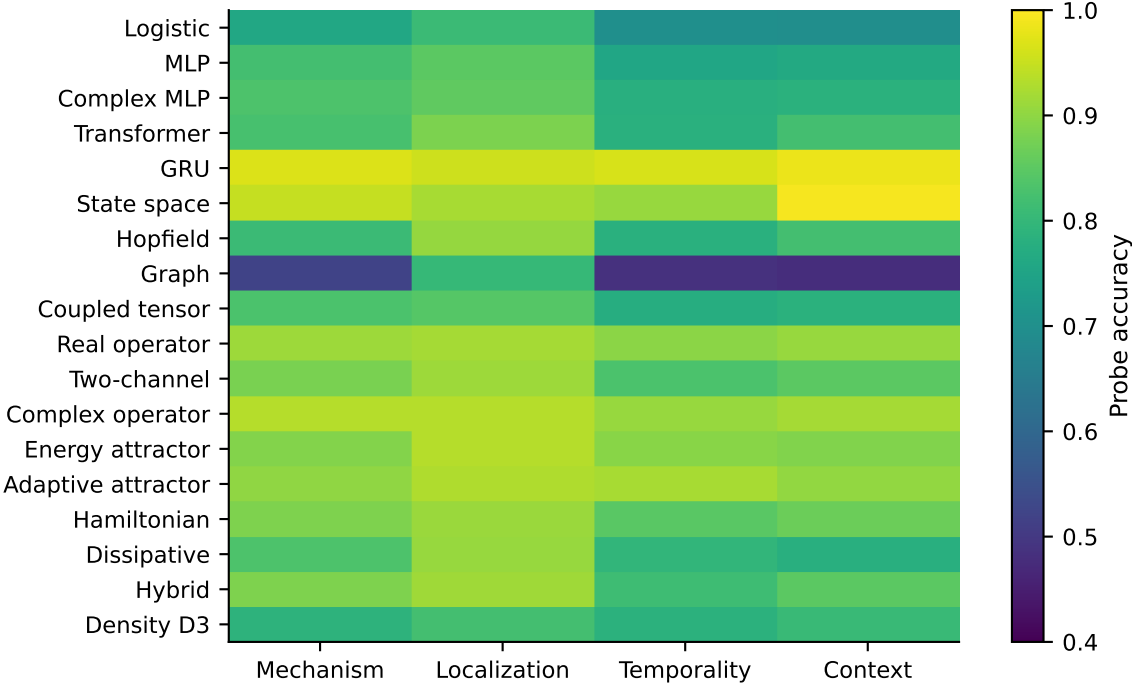
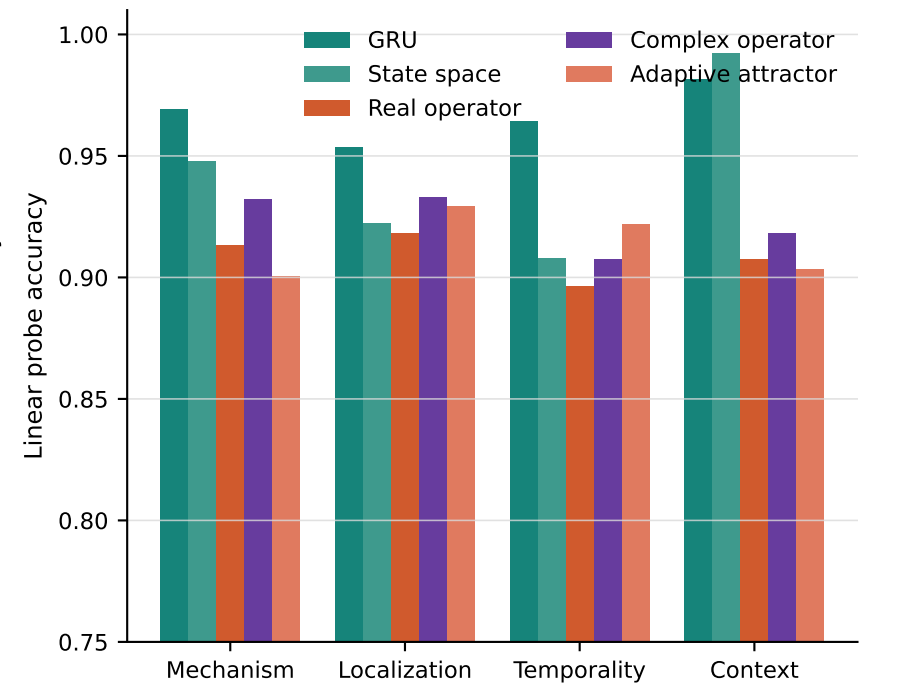


Emergent hierarchical observables

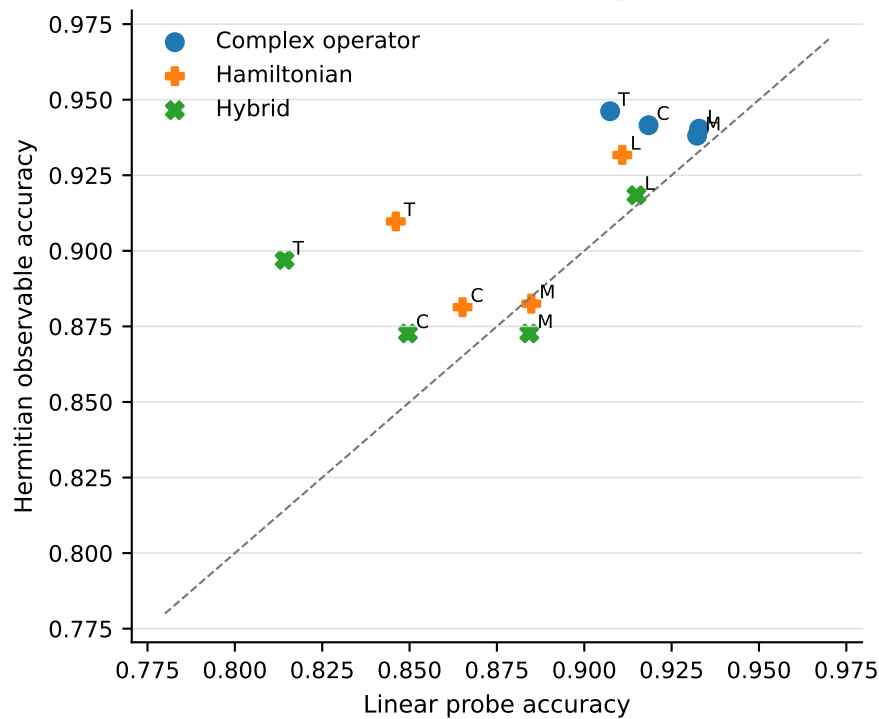
A Linear extractability from frozen states



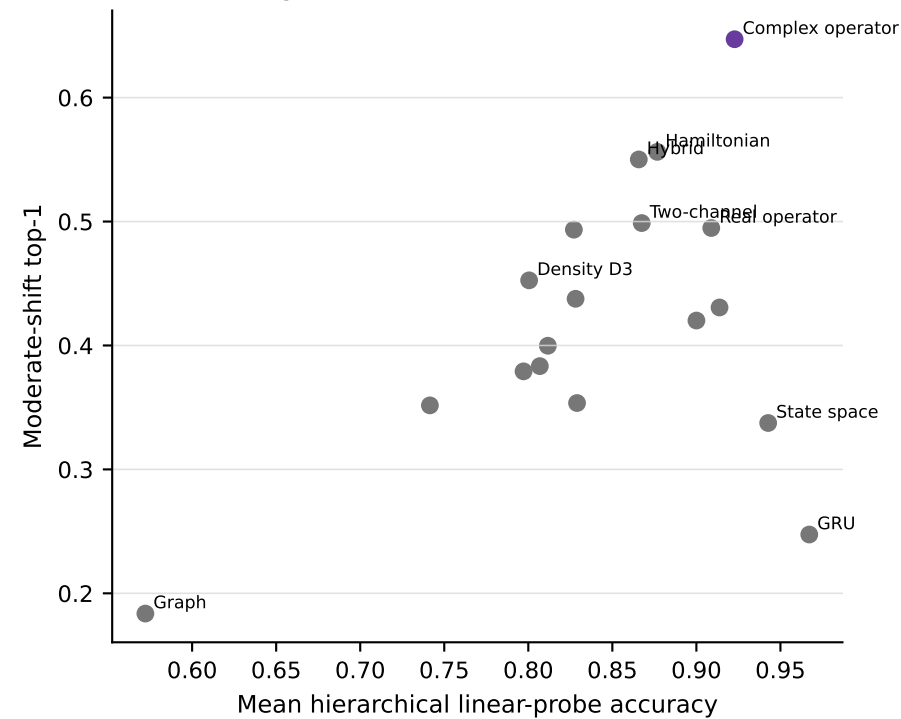
B Strong hierarchy is not unique to complex state



C Quadratic observables can expose structure



D Hierarchy and robustness are distinct ($r=+0.45$)



Probes use frozen QN-000014 representations and training-only validation selection. Means span three model seeds. Source: QN-000019; synthetic data only.